

C++ Project: Quest Archive Manager

A medium-difficulty project that practices core C++ fundamentals without being a typical calculator or CRUD demo.

Project Overview

Build a command-line application that manages a fantasy guild archive. Users can create adventurers, record quests, track rewards, search archives, generate statistics, and save/load data from files.

Required Topics

Functions, arrays or vectors, loops, conditionals, switch statements, input validation, file I/O, strings, structs, sorting, searching, and modular program design.

Core Features

1. Add adventurer. 2. Record completed quest. 3. View all records. 4. Search by name. 5. Sort by gold earned. 6. Generate statistics. 7. Save to file. 8. Load from file. 9. Delete a record.

Data Design

Suggested struct:

Adventurer: name, level, questsCompleted, goldEarned, rank.

Functional Requirements

- 1 Create at least 8 separate functions.
- 2 Store records in an array or vector with a maximum capacity you define.
- 3 Use loops for menus and processing records.
- 4 Use file I/O to persist data between runs.
- 5 Implement at least one sorting algorithm yourself (bubble, selection, insertion).
- 6 Implement a search feature.
- 7 Validate user input.

Suggested Functions

- 1 displayMenu()
- 2 addAdventurer()
- 3 recordQuest()
- 4 searchAdventurer()
- 5 sortByGold()
- 6 showStatistics()
- 7 saveData()
- 8 loadData()
- 9 deleteRecord()

Challenges

- 1 Bronze: Complete all core requirements.
- 2 Silver: Add ranks that automatically update based on quests completed.
- 3 Gold: Add multiple file exports and achievement badges.
- 4 Platinum: Create a quest recommendation system using previous records.

Testing Checklist

- 1 File loads correctly after restart.
- 2 Sorting order is correct.
- 3 Search finds existing records and handles missing ones.
- 4 Invalid input does not crash the program.
- 5 Statistics remain accurate after deletions.

What You Will Learn

This project combines most beginner-to-intermediate C++ fundamentals in one realistic application and naturally prepares you for classes, OOP, STL containers, and larger projects.